Nithin N

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Education

National Institute of Technology Karnataka (NITK), Surathkal Aug. 2024 – May 2028

- **Major:** Electrical & Electronics Engineering **CGPA:** 8.13/10
- **Minor:** Machine Learning **CGPA:** 9.00/10

Experience

Carnegie Mellon University | ***Alzheimer's Disease, Neuroimaging*** July 2026 – Present

Research Project Member *Remote*

- Contributing to Alzheimer's disease neuroimaging research under **Prof. Min Xu**, with guidance from **Xiangrui Zeng (PhD)** at Harvard Medical School, focusing on longitudinal MRI and PET analysis.
- Conducted a comparative analysis of longitudinal brain segmentation and registration methods including **SAMSEG-Long, GeoLongSeg, and V2C-Long**, examining temporal consistency, subject-specific anatomical correspondence, and robustness of longitudinal morphometric measurements.
- Investigated **ADNI longitudinal MRI and PET datasets**, including amyloid, tau, and FDG-PET biomarkers, PET-to-MRI registration, SUVR-based quantification, cortical parcellation, and longitudinal biomarker analysis.
- Developed a research framework for integrating layer-specific cortical segmentation for **supragranular and infragnular layers** with longitudinal MRI-PET measurements to investigate compartment-specific structural atrophy and its relationship with Alzheimer's disease biomarkers.
- Developing an end-to-end longitudinal MRI analysis pipeline, focusing on **image registration and cortical segmentation** to establish anatomically consistent correspondence across repeated scans.

Schindler India | ***Predictive Maintenance, Machine Learning*** Feb. 2026 – May 2026

Machine Learning Researcher *NITK Surathkal*

- Developed a machine learning pipeline for **Remaining Useful Life (RUL) estimation** and predictive maintenance of elevator systems using multi-modal electrical and vibration sensor data for Schindler 5500 lifts.
- Designed a fault diagnosis framework combining **PU Learning, Random Forests, Reinforcement Learning**, and signal processing techniques to address class imbalance, label scarcity, and distribution shift.
- Engineered **18+ time-domain, frequency-domain, and wavelet features** including RMS, kurtosis, crest factor, BPFO harmonics, MCSA sidebands, and wavelet energy for fault classification and health assessment.
- Proposed an adaptive decision engine integrating probabilistic fault prediction with reinforcement learning based threshold optimization for fault severity estimation and maintenance prioritization.

Projects

Quadruped Stand Stabilization | *Reinforcement Learning, MuJoCo, Robotics* May 2026 – Jun 2026

- Developed a terrain-aware **PPO policy** for **Unitree A1** quadruped stand stabilization in MuJoCo using reward shaping, contact-aware constraints, and physics-based simulation.
- Designed a **5-stage curriculum learning pipeline** over **20M simulation steps**, progressively introducing slopes (**5°, 15°, 25°**), rough terrain, mixed environments, and increasing domain randomization complexity.
- Engineered a progressive **DR0–DR7 domain randomization** framework varying friction (**0.5–1.5**), motor strength (**0.80–1.20**), mass (**0.93–1.07**), actuator noise, control delay, and randomized initial poses for robust policy learning.
- Achieved **100%** success on flat terrain and **87.5%** on slopes, maintaining average stable standing durations of **9.37s** and **8.28s**; obtained **50%** and **37.5%** success on rough and stair terrains respectively.

IITM Caterpillar Rover Challenge | *Computer Vision, Edge AI* Oct 2025 – Dec 2025

- Developed an end-to-end **real-time crater detection pipeline**, curating a custom rover imagery dataset and training a YOLO-based detector achieving **0.8219 mAP@50–95**.

- Optimized the deployment pipeline using **model quantization**, preserving prediction confidence between **0.95–0.97** while reducing computational overhead for edge inference.
- Compiled the optimized network into the **Hailo Executable Format (HEF)** and deployed on a **Raspberry Pi 5 + Hailo-8 AI Accelerator**, achieving **24 FPS sustained** and **32 FPS peak** real-time inference.
- Delivered an efficient perception pipeline suitable for **resource-constrained autonomous rover platforms**, balancing detection accuracy, latency, and power efficiency.

XAIQuest: Seeing Beyond Accuracy | *PyTorch, Explainable AI, Computer Vision* Oct 2025 – Nov. 2025

- Benchmarked CNN prediction behavior using **Grad-CAM, Eigen-CAM, Integrated Gradients**, and confidence calibration (**ECE = 0.035**) to analyze feature attribution, prediction reliability, and model failure modes.
- Developed a robustness benchmarking pipeline spanning **8 corruption families** and **78 perturbation settings**, evaluating resilience under Gaussian/Poisson noise, blur, JPEG compression, brightness, contrast, and color perturbations.
- Quantified model degradation under distribution shifts, observing accuracy drops from **89.1% to 21.2%** under severe Gaussian noise and **91.0% to 60.9%** under color perturbations, highlighting dependence on fine-grained spatial features.
- Compared interpretability across multiple CNN architectures, identifying architecture-specific failure patterns and the relationship between confidence calibration and robustness under corrupted inputs.

Leadership & Volunteering

Robotech NITK Aug. 2025 – Present
Core Member *NITK Surathkal*

- Team Captain for the Caterpillar Autonomy Challenge 2027
- Mentored junior members in robotics and reinforcement learning guiding project selection and learning paths.

Web Enthusiasts Club (WEC), NITK October. 2025 – Present
Technical Mentor *NITK Surathkal*

- Conducted technical workshops on **Machine Learning** and **Computer Vision** for undergraduate students.
- Organized hands-on coding sessions and Kaggle competitions to introduce students to practical machine learning workflows and model development.

Achievements

- **Top 10 Teams** — IIM Matrix Startup Program
- **Top 10 / 1922 Teams** — IITM Caterpillar Rover Challenge
- **Top 6 / 120 Teams** — Eco-Ideathon (Rooftop Solar PV Verification)
- **Best Young Minds Award** — Impulse 2026 Signal Processing Hackathon